

Pathmika Weeraratna

Associate Tech Lead — Web & Backend Engineer (Next.js, React & REST APIs)

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Summary

Web and backend-focused engineering leader with nearly five years of experience delivering secure, scalable applications for telecom operators, having grown through every engineering level at Omobio into Associate Tech Lead. Contributed to 25+ client and product engagements – delivery, architecture, and pre-sales solutioning – across 9+ countries in Asia, Africa, and Latin America. Deep hands-on expertise in Next.js/React web platforms, REST API design, payment-gateway integration, and large-scale data/reporting systems (Elasticsearch, PostgreSQL, ClickHouse), alongside a track record of mentoring teams and owning delivery from architecture through production. Deepening technical foundations through an MSc in Artificial Intelligence at the University of Moratuwa. Recognized as Emerging Employee of the Year (2023).

Experience

Associate Tech Lead

Jan 2026 – Present

Omobio (Pvt) Ltd

- Lead, mentor, and scale a cross-functional engineering team of 10+ developers, directing solution design and delivery across multiple product initiatives including centralized reporting platforms.
- Serve as technical subject-matter expert for pre-sales engagements, producing architecture proposals and technical validation that support client acquisition.
- Lead the SOR (Steering of Roaming) centralized reporting platform, architecting full-stack telecom data-reporting solutions from backend pipelines through web dashboards.
- Manage client engagements end-to-end, from requirements gathering through production delivery, debugging and refining software through CI/CD-driven workflows.

Senior Software Engineer

Jan 2024 – Jan 2026

Omobio (Pvt) Ltd

- Led Agile delivery of end-to-end web applications, owning architecture design, effort estimation, and SOW reviews for on-time, client-aligned outcomes.
- Delivered hands-on SQL/CDR reporting engineering for Banglalink's SMSC billing platform (Bangladesh): reverse-engineered field mappings, root-caused report-accuracy defects recovering ~85,000 dropped records, and fixed a 240x query-performance regression (4+ min → ~20 sec) via targeted database indexing.
- Built a production fraud-detection REST service (FastAPI) for a live SMS Firewall deployment, combining embedding-based semantic similarity search with a local-LLM GenAI layer, sustaining ~378 messages/sec at scale.
- Built HOPP, Hutch's Next.js 14 (App Router) web self-care portal — authoring the core auth/locale middleware, a custom Axios API wrapper with unified error handling, and a React Context + useReducer global state store.
- Integrated payment gateway services end-to-end: OAuth2 client-credentials flow, four request types (hold-and-wait, async-poll, status-inquiry, reversal), an 18-scenario error-mapping layer, across Visa, Mastercard, Amex, Alipay, Genie, and Sampath Vishwa payment methods.
- Mentored a team of 10 junior developers, conducted code reviews, and established coding standards, testing practices, and GitHub Actions CI/CD pipelines to improve software reliability.

Software Engineer

Apr 2022 – Jan 2024

Omobio (Pvt) Ltd

- Architected and directed implementation of a centralized web-based telecom reporting platform, processing large-scale datasets via asynchronous task processing (Celery) with containerized deployment (Docker); stack included Python, PostgreSQL, ClickHouse, Elasticsearch, Apache Superset, Prometheus, and Grafana.
- Contributed to Dialog's MDA device-management platform (PHP Yii2 backend, React/Redux-Saga admin console) working within the IMEI blacklist/whitelist (EIR) subsystem, and delivered targeted bug fixes on Dialog's CDR-disclosure platform (PHP Yii2/React).
- Enforced Conventional Commits via Husky + lint-staged + Commitlint git hooks, and standardized Git branching strategy company-wide, later delivered as an internal training program; contributed to ISO 27001 certification readiness.
- Received the Emerging Employee of the Year Award (2023) for outstanding technical contribution.

- Supported frontend and backend feature development for enterprise-grade projects using React.js, Python, and RESTful APIs; transitioned to a full-time Software Engineer role based on strong performance.

Key Projects

- **FirewallBI (Telecom Reporting Platform), 2023–2025:** Architected and led delivery of a multi-tenant BI platform deployed across 8 on-prem operator sites (live references at 1,500 TPS/16.3M subscribers and 500 TPS/3.5M subscribers) — 300+ Logstash JDBC pipelines, custom Kibana RBAC, mutual-TLS Elasticsearch cluster, and a Python reconciliation framework validating data integrity across 200+ pipelines.
- **HOPP Web Self-Care Portal, 2024:** Next.js/TypeScript portal with custom Axios API layer, payment-gateway integration, and a backend-driven CAPTCHA component; delivered under a Hybrid Git Flow model to Hutch's QA environment.
- **SMS Firewall AI/ML Suite — Reporting CMS, 2025–Present:** Built the Next.js management dashboard surfacing fraud-detection outputs and firewall rule controls for the SMS Firewall AI/ML Suite, sitting atop the FastAPI detection backend.
- **MDA (Dialog Device Management Platform):** PHP Yii2 backend with a React/Redux-Saga admin console for carrier-grade IMEI blacklist/whitelist (EIR) device management, integrated with an Erlang/OTP telecom-signaling core.
- **Robi Bangla SMS Opt-In/Opt-Out Platform, 2023:** Reviewed the SRS and contributed solution architecture for a BTRC-mandated language-preference platform (Bangladesh) — React/PHP Yii2/Erlang stack serving 5 user roles with sub-100ms REST APIs.

Additional Experience & Engagements

Beyond web, delivered cross-platform mobile applications in React Native — including a fully custom native alarm engine (~2,700 lines of native Java/Swift) and IAP subscription integrations — and owned Hutch Selfcare for 2+ years, leading its RN version upgrade (0.66.5 → 0.76.5) and 3+ feature change requests. Also gained hands-on AI/ML exposure self-initiating a telecom fraud-detection R&D initiative (semantic embeddings, GenAI, ensemble ML models).

Also contributed to 12+ further engagements, including Dialog's CMDP, RetailHub, WinBack, and DF platforms (Sri Lanka); RoamerSteering and Steering-of-Roaming analytics platforms (2023–Present); and solution proposals for Ncell (Nepal), Grameenphone (Bangladesh), Ethio Telecom (Ethiopia), and BSNL (India).

Areas of Expertise

- **Web:** React.js, Next.js, TypeScript/JavaScript, Redux/Redux-Saga, Tailwind CSS
- **Backend & APIs:** Python, FastAPI, REST API design & development, Microservices, Java (Spring Boot), .NET (ASP.NET Core), PHP (Yii2)
- **Data Engineering:** PostgreSQL, ClickHouse, MariaDB/MySQL, Elasticsearch/ELK, Apache Superset, Apache Airflow, Celery
- **Payments & Integration:** OAuth2, payment-gateway integration, card-network payment methods, SMPP, REST/webhook integration
- **DevOps:** Docker, Git, CI/CD (GitHub Actions, GitLab CI, Jenkins), Linux, Prometheus, Grafana
- **Leadership:** Agile delivery, technical mentoring, pre-sales & client engagement, architecture ownership

Education

Master of Science in Artificial Intelligence (Reading) – University of Moratuwa July 2025 – Present (Expected 2027)

B.Sc. in Information Systems, Second Class Upper Division, GPA 3.44 – University of Colombo School of Computing 2019 – 2022

Diploma in English (Certificate Level), Distinction Pass – Seagis Campus, Nugegoda

2018

Achievements

- Emerging Employee of the Year – Omobio (2023): recognized for exceptional job performance and contribution to key software engineering projects.
- Director's List Awardee – University of Colombo (2021): achieved GPA above 3.7 in first academic year.

References available upon request.